

# Participation Journal

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## Entry 1: (11/14/2021)

Some interesting reflections on the past few chapters:

When we first started discussing sequences of functions, I spent quite a bit of time trying to reconcile the mappings from sets of one size to sets of another size. I thought of Galileo's paradox which I had learned about in my Discrete Math class—that is,  $f: \mathbb{N} \rightarrow \mathbb{N}$ , and say the function  $f(x) = x^2$  for  $x \in \mathbb{N}$ , then the function maps to the set  $\{1, 4, 9, 16, \dots\}$ . As you can see, there is a one-to-one mapping between these two sets by the function, but the supposed paradox comes when we look at the cardinality of the sets and the fact that the codomain is a proper subset of the domain. After some discussion with people in the math center, regarding the sequences of functions, and I learned that we select an  $x$  for the purposes of the sequence at which point we're simply dealing with a normal sequence in terms of  $n$ .

During this time, however, I watched several videos on measure theory and came across a discussion about how  $0 \cdot \text{infinity}$  is often defined to be 0, which is intuitive as you have 0 infinities. Furthermore, I read some interesting thoughts about how some consider it to be an indeterminate form depending on the context, such as when taking a limit. More interesting still is that in combinatorics  $0^0$  is defined to be 1. So now we have roughly the same concept dealt with differently in different contexts. In measure theory, as I understand it, they are interested in finding a length of some subset—say of the real numbers. Or in  $\mathbb{R}^2$  they are interested in finding the area of some subspace. This idea got me thinking about the rectangle area we find when taking a Riemann sum. Consider the measure theory concept of length for some set of real numbers  $(b-a)$ .

Now consider a partition of this set which will be used for a Riemann sum. And suppose we have the length between any 'two consecutive' terms of the partition, as they may be arranged on the real line such that  $x_n < x_{n+1}$ . Now the length between any two terms is defined to be  $x_{n+1} - x_n$ . Suppose we call a function  $L(x)$  to simply be this length for a single index of a partition. The concept I am going for is that we have some kind of way of viewing these things as they trend (as we take finer and finer partitions). Clearly the consecutive pairwise length will be drawing close to 0 for more and more elements of the set that partitions the length of the line. Now consider a function 'over' this real line. We want to find the area under this line. At every element of the partition, the function is defined to be  $f(x_n)$ . So, any rectangle formed is defined to be  $(x_{n+1} - x_n) \cdot f(x_n)$  (either  $x_n$  or  $x_{n+1}$ ). Now suppose this is another function.

The limit of this function as the partition set grows arbitrarily large goes to 0, you'd think. So to come full circle, my point here is this: Suppose the sum can be viewed as  $n \cdot (x_{n+1} - x_n) \cdot f(x_n)$  (either  $f(x_n)$  or  $f(x_{n+1})$ ). As  $n \rightarrow \text{infinity}$ , we get something along the lines of  $0 \cdot \text{infinity}$ . Which, in measure theory, is supposedly defined to be 0, but since we're taking the limit it may be considered an indeterminate form. So how might we deal with this case? Given any partition of the set,  $n \cdot (x_{n+1} - x_n)$  should 'cancel' to be just the normal length of that subset  $(b-a)$ . However, it is an interesting thought.

I realize in retrospect that this was incredibly informal and some of the concepts I am trying to convey are likely completely false when recast in a rigorous light, which they ought to be.